Subject: Artificial Intelligence (417)
Total Marks: 30
Time: 1 Hour

SECTION A – MULTIPLE CHOICE QUESTIONS (1 Mark Each)

Q1. An AI hiring tool is designed to review text-based resumes and rank candidates. The developers intentionally deleted all references to gender and race from the training data. However, the AI still ends up rejecting 90% of female applicants because it subtly learned to penalize gaps in employment (which disproportionately affected women taking maternity leave). What specific ethical failure does this scenario demonstrate?
A) Lack of Accountability
B) Algorithmic Bias and Unfairness
C) Violation of Data Privacy
D) High Variance

Q2. An AI model designed to detect deepfake videos achieves 100% accuracy during its training phase. However, when deployed in the real world to evaluate brand-new, unseen videos, its accuracy drops to 45%. Mathematically, what does this vast discrepancy indicate about the model?
A) The model has perfectly generalized the data.
B) The model is underfitting due to low complexity.
C) The model has overfit and memorized the training data's specific noise.
D) The model is executing an Unsupervised Learning clustering approach.

Q3. In a dataset tracking the time it takes a facial recognition AI to process an image, 95% of the images are processed in exactly 0.5 seconds. However, 5% of the images are heavily corrupted by bad lighting and take 45 seconds to process, creating a very long tail stretching to the right side of the distribution graph. Which statistical measure will be most heavily distorted by these corrupted images?
A) The Median
B) The Mean
C) The Mode
D) The True Negatives

Q4. A legal software company wants an AI to scan through 10,000 pages of corporate contracts and automatically group clauses that have similar legal phrasing, even though the developers haven't defined any specific categories (like "Liability" or "Payment") in advance. Which learning model must the system use?

A) Supervised Classification
B) Rule-Based Expert System
C) Unsupervised Clustering
D) Reinforcement Learning

Q5. A team of engineers completely abandons an Artificial Neural Network (ANN) and switches to a Convolutional Neural Network (CNN) to process the video feed for a self-driving car's stop-sign detection. What is the fundamental technical reason for this switch?
A) CNNs do not require any training data to function.
B) CNNs are designed to process sequential text, while ANNs only process numbers.
C) CNNs can process the complex spatial hierarchies, edges, and textures in visual images far better than ANNs.
D) CNNs completely bypass the need for an Input Layer and Output Layer.

Q6. A medical AI scans blood samples for a highly contagious, deadly pathogen. If the AI misses a sick patient, the patient will spread the disease. If the AI falsely flags a healthy patient, they simply take a harmless secondary blood test. To maximize public safety, which evaluation metric must the developers prioritize?
A) Maximizing Precision to avoid False Positives.
B) Maximizing Recall to minimize False Negatives.
C) Maximizing True Negatives to ensure healthy people are ignored.
D) Maximizing Overall Accuracy regardless of class imbalance.

Q7. A user in the Orange Data Mining platform has successfully trained a classification model. They now want to visually compare the model's predicted labels against the actual labels to see exactly where the AI is making mistakes. Which specific widget category must they open to find the appropriate tool?
A) Transform Widgets
B) Visualize Widgets
C) Evaluate Widgets
D) Data Reduction Widgets

Q8. A smart home system uses a microphone to capture a user saying, "It's too dark in here." The system translates the audio to text, understands the underlying intent of the complaint, and then turns on the living room lights. Which AI domain is responsible for extracting the intent from the user's spoken phrase?
A) Natural Language Processing (NLP)
B) Computer Vision (CV)
C) Reinforcement Learning (RL)
D) Descriptive Statistics

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each)

Q9. An AI researcher is evaluating a dataset of 1,000 credit card transactions where only 2 are fraudulent. Explain mathematically why calculating the "Mean" to find the average transaction amount might be perfectly valid, but relying on the "Accuracy" metric to evaluate the AI's fraud detection model is dangerously flawed.

Q10. A regional hospital uses a proprietary No-Code AI platform to predict patient heart attacks. The platform does not allow users to see or edit the underlying code or algorithms. Explain the inherent tension between the ethical principle of "Transparency" and the use of this specific No-Code tool in a high-stakes healthcare setting.

Q11. During the evaluation of an AI-powered email spam filter, distinguish between the practical, real-world impact of a True Negative (TN) versus a False Negative (FN) on the human user checking their inbox.

Q12. An AI agent is learning to manage the power grid of a city through Reinforcement Learning. Explain how the concepts of "reward" and "penalty" would be specifically defined and mathematically applied in this high-stakes scenario to ensure the grid doesn't crash.

Q13. A university wants software to check if students are plagiarizing essays. Contrast how a "Rule-Based Approach" would detect plagiarism versus how a "Learning-Based Approach" (using AI) would accomplish the same task.

SECTION C: LONG ANSWER / APPLICATION QUESTIONS (4 Marks Each)

Q14. An aerospace engineering firm uses an AI model to detect microscopic manufacturing defects in airplane engines via X-ray scans. Out of 500 engines tested last month, the system generated the following results:

- The model correctly flagged 40 defective engines.
- The model correctly cleared 420 perfectly safe engines.
- The model incorrectly flagged 30 safe engines as defective.
- The model completely missed 10 defective engines, passing them as safe for flight.

a) Extract and identify the specific numerical values of TP, TN, FP, and FN from the scenario text. *(1 mark)*
b) Calculate the **Precision** and the **Recall** of this AI model. (Show your formulas). *(2 marks)*
c) The aviation authority demands that *absolutely no* defective engine ever makes it onto an airplane.

Based on your calculations, explain which specific metric is failing to meet this critical safety demand and why. *(1 mark)*

Q15. An agricultural technology company is designing an autonomous robotic tractor to optimize farming.

a) The tractor uses a camera to scan the field and distinguish between healthy crops and harmful weeds. Justify why the development team must utilize a Convolutional Neural Network (CNN) instead of a standard Artificial Neural Network (ANN) for this specific visual task. *(2 marks)*

b) Once the tractor scans the field, it uploads the massive amounts of unlabelled soil moisture data to a central server. The server automatically finds hidden patterns and groups the field into distinct moisture zones without any prior human categorization. Identify this specific machine learning approach and explain its fundamental mechanism. *(2 marks)*

Q16. A state government wants to deploy an AI system to predict which specific districts will face severe water shortages next summer so they can proactively route water trucks.

a) In the "Problem Scoping" stage of the AI Project Cycle, identify who the primary stakeholders are and define the exact problem block. *(1 mark)*

b) During "Data Acquisition," the team gathers 20 years of historical rainfall data, but notices an entire year's worth of data is completely missing due to a past sensor failure. Identify the specific data preparation process used to estimate and replace this missing data without discarding the whole dataset. *(1 mark)*

c) During "Data Exploration," the team plots the daily water usage of the citizens on a graph and finds it has a severe right-skewed (positive skew) distribution. Explain what this visual shape implies about the water usage habits of the population (where does the majority sit versus the extreme outliers?). *(1 mark)*

d) After deployment, the government officials blindly trust the AI's predictions and refuse to send water trucks to a clearly drying district because "the computer didn't flag it." Identify this specific psychological phenomenon and explain how it undermines human decision-making. *(1 mark)*